T063-PS-004: Payroll, Attendance and Leave Engine

Design document. Team T063, section CSD. 10 SECONDS capstone, PS-004 (Python 3.10+).

1. Problem and approach

The company needs a monthly payroll for 200 employees driven by text files: an employee master, daily attendance, a holiday calendar, statutory tables and a command file that records every event (leave applications, approvals, revisions, exits, payroll runs, year-end). There is no UI. The program reads the files, replays the command file strictly in order, and writes CSV outputs (payslips, leave ledger, balances, full-and-final statements, command log).

We split the system into two halves that meet at one object. The leave and attendance half decides how many days each employee is paid for. The payroll half turns those days and the CTC into money. Every business rule is numbered BR-01 to BR-34 and each is implemented in one function and covered by named tests (section 4).

2. Architecture

Module	Responsibility
config.py	Every rule number (12, 30, 45, 15th, 60 days, 5 working days) in one place.
loaders.py, models.py	Read the CSVs; clean duplicate rows, unknown employees, invalid statuses and log each as a data issue.
calendar_utils.py	Weekly-offs, holidays, working days, add N working days.
accrual.py	Monthly CL/SL credit, joiner rule, EL by quarter, year-end carry-forward and lapse.
leave_rules.py, leave_requests.py	Day counting (sandwich), validity rules, approval chain, auto-approval, balance charging.
attendance.py	Daily statuses plus leave cover become payable days (MonthlyAttendanceSummary).
payroll.py	Earnings, PF, PT, TDS, payslip, full-and-final, revisions.
commands.py, pipeline.py	Parse and replay commands.txt; connect the two halves; write all outputs.

Data flow: commands.txt is read line by line. Leave commands update the leave engine. RUN_PAYROLL loads that month's attendance, asks the leave engine which days are covered by approved leave, builds a MonthlyAttendanceSummary per employee, and hands the summaries to the payroll engine. EXIT, REVISE_CTC, REGIME_CHANGE and FNF_RELEASE go to the payroll engine through a handler table.

The handshake object is MonthlyAttendanceSummary: calendar_days, payable_days, lop_days, unauthorised_days, half_day_unpaid, not_employed_days and payable_by_date (the payable amount of each calendar day, which lets payroll split a month at a salary revision).

3. Key design decisions

Time follows the command file. Before each command the accrual clock moves to that command's date and credits every month-start reached. A month of the next leave year is not credited until YEAR_END has run, otherwise January's CL would be credited and then wiped by the year-end lapse.

Balances are reserved at application time. A request takes its days from the balance when applied and gives them back on reject or cancel. If the balance is short, the excess days become LOP with a flag. Days that fall in a month not yet accrued are deferred and charged when that month is credited.

Sandwich rule. For EL and LOP, weekly-offs and holidays between the first and last working leave day count as leave. CL, SL and comp-off count working days only. Maternity counts every calendar day.

Money is Decimal. Floats would break component-by-component matching. Each earning component is rounded half-up to the paisa and gross is the sum of the rounded components.

Rules in one place, assumptions in one log. Every interpretation is recorded in docs/decision-log.md with a pointer to the code and a status (Brief, Assumed or ASK the trainers).

4. Business rule register

ID	Rule	Implemented in	Tests
BR-01	CL 1/month on the 1st, max 12, lapses at year-end	accrual.accrue_month / year_end	test_accrual
BR-02	SL 1/month, max 12 accrued, carry-forward up to 30	accrual.accrue_month / year_end	test_accrual
BR-03	EL 1.5/month credited quarterly in arrears, carry up to 45	accrual.credit_el_quarter / year_end	test_accrual
BR-04	Joiner accrues for the joining month only if joined on/before the 15th	accrual.accrual_eligible	test_accrual
BR-05	Year-end carry-forward / lapse	accrual.year_end	test_accrual, test_leave_requests
BR-06	SL over 3 consecutive days needs a medical flag	leave_requests._validate	test_leave_requests
BR-07	CL cannot be clubbed with EL	leave_rules.is_clubbed	test_leave_rules, test_leave_requests
BR-08	EL needs 7 days' notice	leave_requests._validate	test_leave_requests
BR-09	Comp-off expires in 60 days	models.CompOffCredit, balances.CompOffLedger	test_loaders_and_compoff
BR-10	Sandwich rule for EL and LOP, not CL/SL	leave_rules.counted_days	test_leave_rules
BR-11	Insufficient balance converts to LOP with a flag	leave_requests._charge_balance	test_leave_requests
BR-12	Approval chain by grade (G1-2 manager; G3-4 manager+HR; G5 HR)	leave_rules.required_roles, leave_requests.approve	test_leave_requests
BR-13	Auto-approve after 5 working days, except LOP	leave_requests.sweep_auto_approvals	test_leave_requests
BR-14	Attendance statuses P/A/HD/H/WO/WFH imported	loaders.load_attendance	test_loaders_and_compoff
BR-15	Payable days = calendar - LOP - unauthorised; HD 0.5; WFH present	attendance.summarise_month	test_attendance
BR-16	Basic = 40% of CTC	payroll.components	test_payroll
BR-17	HRA 50% of basic metro / 40% other	payroll.components	test_payroll
BR-18	Special allowance balances CTC	payroll.components	test_payroll
BR-19	Earnings pro-rated by payable / calendar days	payroll.earnings	test_payroll
BR-20	PF 12% of basic, employee and employer	payroll.provident_fund	test_payroll
BR-21	Statutory PF basic capped at Rs 15,000	payroll.provident_fund	test_payroll
BR-22	Opt-out of PF on the excess allowed	payroll.provident_fund	test_payroll
BR-23	Karnataka PT: nil up to 25,000, 200 above	payroll.professional_tax	test_payroll
BR-24	February PT Rs 300	payroll.professional_tax	test_payroll
BR-25	Other states' PT from the table	payroll.professional_tax	test_payroll
BR-26	TDS = (projected tax - TDS paid) / remaining months, monthly	payroll.tds_for_month	test_payroll
BR-27	Regime switch mid-year rejected	payroll._h_regime	test_payroll, test_pipeline
BR-28	Mid-month joiners and exits proportional	attendance, payroll.earnings	test_payroll, test_pipeline
BR-29	Revision effective any date (split by days)	payroll.ctc_on / earnings	test_payroll
BR-30	F&F;: pay to last working day	payroll._full_and_final	test_payroll
BR-31	EL encashment at basic/30	payroll.run_month	test_payroll
BR-32	Notice shortfall recovery at gross/30	payroll._full_and_final	test_payroll
BR-33	Asset deductions	payroll._full_and_final	test_payroll
BR-34	F&F; release 45 days after LWD	payroll._full_and_final, _h_release	test_payroll

5. Edge cases

Edge case	How it is handled	Tests
Leave spanning a month boundary	Days are counted once; the daily cover map splits them by month, so January's run sees 4 days and February's 3.	test_leave_rules, test_leave_requests
Leave spanning year end with carry-forward	Days in a month not yet accrued are deferred. Old-year days are charged first, YEAR_END credits Q4 EL and applies the cap, then new-year days are charged.	test_leave_requests (year-end tests)
Joiner on the 16th	accrual_eligible gives nothing for December; attendance rows start on the 16th, so payable = 16 of 31 calendar days.	test_accrual, test_payroll, test_pipeline
Exit during notice with negative balance	Balance never goes below zero (excess becomes LOP). Notice shortfall and asset recovery can make the F&F; net negative.	test_payroll
Revision on the 1st vs mid-month	ctc_on() is evaluated per day, so a revision on the 1st changes the whole month and a mid-month revision splits it.	test_payroll
February PT	Slab lookup returns february_pt in month 2 (Rs 300 in Karnataka).	test_payroll
Regime switch mid-year	REGIME_CHANGE always raises CommandRejected (BR-27); the command log shows REJECTED.	test_payroll, test_pipeline
Maternity overlapping a payroll run	Maternity is a paid leave covering every calendar day; absent days under it are payable, so the run pays in full.	test_attendance, test_leave_requests

6. Worked payslip: CTC to net

Employee: G4, metro (Bengaluru), Karnataka, new regime, CTC Rs 24,00,000, PF on full basic. December 2026 (31 days), one unauthorised absence on 14 December. Rs 16,00,000 gross and Rs 1,95,000 TDS were already paid in April to November. The slabs are the FY 2026-27 new-regime slabs; in a real run they come from it_slabs.csv.

Step	Calculation	Result (Rs)
Monthly CTC	2,400,000 / 12	200,000.00
Basic (40%)	200,000 x 40%	80,000.00
HRA (metro 50% of basic)	80,000 x 50%	40,000.00
Special allowance	200,000 - 80,000 - 40,000	80,000.00
Payable days	31 calendar days - 1 unauthorised	30
Pro-rating	each component x 30/31	basic 77419.35, HRA 38709.68, special 77419.35
Gross	sum of the three	193548.38
PF employee (12% of basic)	12% x 77419.35	9290.32
Professional tax	Karnataka, gross above 25,000	200
Projected annual gross	1,600,000 + 193548.38 + 3 x 200,000	2393548.38
Taxable income	projected - 75,000 standard deduction	2318548.38
Annual tax	slabs + 4% cess (no rebate: above 12 lakh)	290822.58
TDS this month	(290822.58 - 195,000) / 4 remaining months	23955.65
Total deductions	PF + PT + TDS	33445.97
Net pay	gross - deductions	**160102.41**
Employer cost	gross + employer PF 9290.32	202838.70

Slab check on taxable income 2318548.38: 4-8 lakh at 5% = 20,000; 8-12 lakh at 10% = 40,000; 12-16 lakh at 15% = 60,000; 16-20 lakh at 20% = 80,000; the remaining 318548.38 above 20 lakh at 25% = 79637.10; total 279637.10; with 4% cess 290822.58.

7. Validation and data cleaning

Attendance rows are checked against the employee master and the allowed statuses. A duplicate (employee, date) row is counted once; a row for an unknown employee is skipped; each is logged as a data issue and printed at the end of the run. A missing column in an input file stops the run with a clear message. Every leave request is validated in a fixed order and the first failure sets REJECTED with its rule id. Every command is logged with a status; one bad line never stops the run. MonthlyAttendanceSummary carries a built-in identity check: payable + LOP + unauthorised + half-day remainder + not-employed days always equals the calendar days.

8. Testing

119 pytest tests across eight files: calendar, accrual and year-end, attendance and payable days, loaders and comp-off, leave rules, the leave request lifecycle, payroll arithmetic, and whole-engine runs that start from CSV files and check the output files. The year-end test reproduces the README scenario (EL 29 Dec to 4 Jan with a balance of 40: 37, then 41.5 after the Q4 credit, then 37.5). The TDS calculation was cross-checked against published FY 2026-27 figures (Rs 15 lakh salary, about Rs 97,500 including cess).

9. Assumptions, limits and future work

The full list of interpretations is in docs/decision-log.md; the main open questions are accrual timing (inside or before RUN_PAYROLL), whether payslip lop_days includes unauthorised absences, and the basis of gross/30 in full-and-final. TDS ignores surcharge, marginal relief, HRA exemption and professional-tax deduction. Arrears for late salary revisions and per-location leave policies from JSON (the stretch goals) are not implemented; a late revision is stored and a warning is printed.